WORKED SOLUTION

STEP II 2022, Question 12

Probability and Statistics • Mean, Median and Mode

STEP II 2022, Question 12

Read the question before you go on. It is in the STEP II paper in OCR's download of the 2022 STEP papers (a zip file):

www.ocr.org.uk/Images/696587-2022-past-papers.zip

The key idea

Everything flows from the cumulative distribution function F, which here is a simple polynomial. Since F is increasing on $[0,1]$, a number a lies below the median exactly when $\mathrm{F}(a) < \frac{1}{2}$. So to compare the mean or the mode with the median, we never need to find the median itself: we just evaluate F at the mean or the mode and compare with $\frac{1}{2}$.

Solution

Part (i)

Since the total probability is 1,

$$1 = k\int_0^1 \left(x^n - x^{n+1}\right)\mathrm{d}x = k\left(\frac{1}{n+1} - \frac{1}{n+2}\right) = \frac{k}{(n+1)(n+2)},$$

so $k = (n+1)(n+2)$, as required. Then

$$\mu = k\int_0^1 \left(x^{n+1} - x^{n+2}\right)\mathrm{d}x = (n+1)(n+2)\cdot\frac{1}{(n+2)(n+3)} = \frac{n+1}{n+3}.$$

Part (ii)

For $0 \le x \le 1$,

$$\mathrm{F}(x) = (n+1)(n+2)\left(\frac{x^{n+1}}{n+1} - \frac{x^{n+2}}{n+2}\right) = x^{n+1}((n+2) - (n+1)x).$$

Because $\mathrm{f} > 0$ on $(0,1)$, F is strictly increasing there, and the median m is the unique point with $\mathrm{F}(m) = \frac{1}{2}$. Hence

$$\mu < m \iff \mathrm{F}(\mu) < \tfrac{1}{2}.$$

Now with $\mu = \frac{n+1}{n+3}$,

$$(n+2) - (n+1)\mu = \frac{(n+2)(n+3) - (n+1)^2}{n+3} = \frac{3n+5}{n+3},$$

so

$$\mathrm{F}(\mu) = \left(\frac{n+1}{n+3}\right)^{n+1} \frac{3n+5}{n+3}.$$

The condition $\mathrm{F}(\mu) < \frac{1}{2}$ therefore becomes, after multiplying by the positive quantity $2\left(\frac{n+3}{n+1}\right)^{n+1}$,

$$\frac{2(3n+5)}{n+3} < \left(\frac{n+3}{n+1}\right)^{n+1}.$$

Since $\frac{2(3n+5)}{n+3} = \frac{6(n+3)-8}{n+3} = 6 - \frac{8}{n+3}$ and $\frac{n+3}{n+1} = 1 + \frac{2}{n+1}$, this is exactly the given inequality. So μ is less than the median whenever that inequality holds.

To prove the inequality, write $N = n+1$, so $N \geq 3$ is an integer. Every term of the binomial expansion of $\left(1 + \frac{2}{N}\right)^N$ is positive, and there are $N+1 \geq 4$ terms, so the sum is at least its first four terms:

$$\begin{aligned}\left(1 + \frac{2}{N}\right)^N &\geq 1 + N \cdot \frac{2}{N} + \frac{N(N-1)}{2} \cdot \frac{4}{N^2} + \frac{N(N-1)(N-2)}{6} \cdot \frac{8}{N^3} \\ &= 3 + 2\left(1 - \frac{1}{N}\right) + \frac{4}{3}\left(1 - \frac{3}{N} + \frac{2}{N^2}\right) \\ &= \frac{19}{3} - \frac{6}{N} + \frac{8}{3N^2}.\end{aligned}$$

Ask yourself The four-term sum is only a lower bound. How do we show it beats $6 - \frac{8}{N+2}$?

We subtract and clear the positive denominator $3N^2(N+2)$:

$$\begin{aligned}&3N^2(N+2)\left[\frac{19}{3} - \frac{6}{N} + \frac{8}{3N^2} - 6 + \frac{8}{N+2}\right] \\ &\quad = N^2(N+2) - 18N(N+2) + 8(N+2) + 24N^2 \\ &\quad = N^3 + 8N^2 - 28N + 16.\end{aligned}$$

For $N \geq 3$ we have $N^2 + 8N - 28 \geq 9 + 24 - 28 = 5 > 0$, so $N^3 + 8N^2 - 28N + 16 = N(N^2 + 8N - 28) + 16 > 0$. Hence

$$6 - \frac{8}{n+3} < \frac{19}{3} - \frac{6}{N} + \frac{8}{3N^2} \leq \left(1 + \frac{2}{n+1}\right)^{n+1},$$

the given inequality holds for every integer $n > 1$, and so the median of X is greater than μ.

Part (iii)

On $(0,1)$,

$$\mathrm{f}'(x) = k\left(nx^{n-1} - (n+1)x^n\right) = kx^{n-1}(n-(n+1)x),$$

which is positive for $x < \frac{n}{n+1}$ and negative for $x > \frac{n}{n+1}$. So f rises then falls, and the mode is $M = \frac{n}{n+1}$.

As in part (ii), $M > m$ if and only if $\mathrm{F}(M) > \frac{1}{2}$. Since $(n+2) - (n+1)M = (n+2) - n = 2$,

$$\mathrm{F}(M) = 2\left(\frac{n}{n+1}\right)^{n+1},$$

so we need $\left(\frac{n}{n+1}\right)^{n+1} > \frac{1}{4}$, that is,

$$\left(1+\frac{1}{n}\right)^{n+1} < 4.$$

? **Ask yourself** How does the given fact about $\left(1+\frac{1}{x}\right)^{x+1}$ help, when n could be anything from 2 upwards?

Because the function is decreasing for positive x, its value at any integer $n \geq 2$ is at most its value at $x = 2$:

$$\left(1+\frac{1}{n}\right)^{n+1} \leq \left(\frac{3}{2}\right)^3 = \frac{27}{8} < 4.$$

Hence $\mathrm{F}(M) > \frac{1}{2}$, and the mode of X is greater than its median.

Putting the parts together, mean $<$ median $<$ mode for every $n > 1$.

Checks

- Take $n = 2$: the mean is $\frac{3}{5} = 0.6$ and the mode is $\frac{2}{3} \approx 0.667$. Solving $4x^3 - 3x^4 = \frac{1}{2}$ numerically gives median ≈ 0.6143, neatly in between. Also $\mathrm{F}(0.6) = 0.4752 < \frac{1}{2}$ and $\mathrm{F}(\frac{2}{3}) = \frac{16}{27} \approx 0.593 > \frac{1}{2}$.
- At $n = 2$ ($N = 3$) the expansion has exactly four terms, so the lower bound is an equality: both sides equal $\frac{125}{27} \approx 4.63$, comfortably above $6 - \frac{8}{5} = 4.4$.
- The ordering mean $<$ median $<$ mode was confirmed by computer for every n from 2 to 79.

Common slips

- Trying to find the median explicitly: the equation $x^{n+1}(n+2-(n+1)x) = \frac{1}{2}$ has no nice solution, and you don't need one.

- Getting the direction wrong: $\mu < m$ corresponds to $\mathrm{F}(\mu) < \frac{1}{2}$, because F is increasing.
- Forgetting to say why dropping the later binomial terms is allowed (they are all positive).

Beyond STEP

This density is a Beta distribution, $\mathrm{Beta}(n+1, 2)$, one of the most important families on $[0,1]$ and central to Bayesian statistics. The ordering mean $<$ median $<$ mode is typical of a distribution with a long tail to the left (negative skew), though it is not a universal law, which is why it had to be proved here. The decreasing function in part (iii) tends to e as $x \to \infty$, a classic limit that you will meet properly in analysis. Blitzstein and Hwang, *Introduction to Probability*, has an excellent treatment of the Beta distribution.

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